



Ref: SMTTP/Chemistry/2022-23/OFC NOTE/268266

Date:

12/07/2022(11:49:22)

Subject: LMI PREVENTION OF DAMAGE TO GENERATORS DUE TO MOISTURE IN HYDROGEN

Ref: NTPC Simhadri/CHEM/LMI-76

Date: 12.07.2022

SUB: Review of LMI-Prevention of damage to generators due to Moisture in hydrogen

The following LMI pertaining to O&M (CHEM) section has been reviewed and no Changes Incorporated.

Sl.No	Title of LMI & Revision details	Changes/Revisions Made	Remarks
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1	Prevention of damage to generators due to moisture in hydrogen.	LMI has been reviewed no changes Incorporated.	Next review shall be in July-2024 or if OGN is revised, whichever is earlier.
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LMI/OGN/OPS/SYST/009

Rev.4, 21.07.2020

Submitted for approval please.

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PREVENTION OF DAMAGE TO GENERATORS DUE TO MOISTURE IN HYDROGEN

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1.0 INTRODUCTION:

Hydrogen is used as the cooling medium in Simhadri TPS generators as its high heat transfer co-efficient and low density result in very good heat transfer property with low windage & friction losses. However, presence of moisture more than a particular proportion of total volume in Hydrogen gas is detrimental to insulations of windings and rotor end retaining rings. Any condensation of moisture may result in electrical flashovers particularly on end windings, stress corrosion cracking of austenitic rotor retaining rings and appearance of rotor earth faults.

2.0 SUPERSEDED DOCUMENTS:

LMI/OGN/OPS/SYST/009 Issue: 01, Revision: 04

3.0 SCOPE:

These guidelines are applicable to Simhadri 500MW generators. The scope of these guidelines is to cover basic principles and recommendations keeping in view of manufacturer's recommendations and the cooling system layouts.

4.0 BASIC TERMS AND DEFINITIONS:

Basic Terms & Definitions are given in Annexure - I

5.0 SPECIFIC INSTRUCTIONS TO BE FOLLOWED :

Sl no	Activity	Responsibility
5.1	Operating limits for cold gas temperature are set in such a way that an extreme possible condition of low temperature of cold gas shall not result in condensation of moisture. This is achieved either by regulating cooling water flow through the gas coolers or by decreasing dew point of gas by reducing the quantity of moisture in the gas itself. Normal operating cold gas temperature is 37 °C with a maximum of 55 °C. High alarm is at 55°C and very high alarm is at 60°C. Generator protection is given above 60°C . It is to be ensured that the dew point of the gas-vapor in the casing shall be at least 20°C lower than the lowest cold gas temperature and cold gas temperature limits are maintained.	SCE/Unit Control Engineer

Sl no	Activity	Responsibility
5.2	It is necessary to maintain a temperature differential between cold primary water temperature and cold gas temperature with primary water temperature on higher side so that moisture, if any, in the gas does not condense on any solid surfaces such as stator winding, cooling water hoses, end windings / winding connections. Normal operating differential between cold primary water temperature and cold gas temperature should be more than 5°C and low alarm is at 3°C and very low alarm is at 1°C.	SCE/Unit Control Engineer
5.3	Drier units help in bringing down the moisture level in casing hydrogen. These driers can be put into service only when machine is at 3000 rpm and cooling fan differential forces the hydrogen through the drier. It is therefore, important to maintain hydrogen pressure in the casing during shut down and both vapor extractors of seal oil system shall be kept running so that the chance of moisture transfer from seal oil is minimized.	SCE/Unit Control Engineer
5.4	H ₂ Driers provided are refrigeration Type. The moisture in the hydrogen condenses through cooling and converts in water form in condensate chamber. It is to be ensured that all the moisture is removed by proper draining within 2 to 3 days after checking moisture level through looking glass window. It should be ensured that both the drain valves are closed properly after removal of water and only water is drained out	SCE/Unit Control Engineer
5.5	Regular monitoring is required to be done of casing drains (LLDS), although an alarm is provided when the level in "LLD" has built up. LLDs should be checked at least once in every shift and recorded. If water is present and it is continuous the unit should be de energized, the cause of water leakage identified and attended.	SCE/Unit Control Engineer
5.6	Before energizing the machine after a prolonged shut down, Dew point shall be measured. In case dew point is not at least 20deg C below cold gas temperature, the machine shall not be energized. The moisture level shall be brought down to required level by putting driers in service. The specific value of dew point is to be recorded and clearance for exciting the generator is to be obtained from Chemistry and Electrical sections.	Shift Chemist & SCE/Unit Control Engineer/

Sl no	Activity	Responsibility
5.7	Whenever the unit is to be brought back after a shutdown of moderate duration or after filling of generator with fresh Hydrogen gas, the dew point shall be measured before exciting the generator. The dew point reading shall be recorded in the control room Chemistry log book	Shift Chemist
5.8	In cases, when casing hydrogen dew point has increased to 10 deg C immediate action shall be taken to improve it by using drier and purging of casing hydrogen with fresh gas till the dew point achieves normal value.	SCE/Unit Control Engineer
5.9	There is on line measurement at drier inlet and drier outlet temperatures to assess effectiveness of drier. These temperatures are to be monitored and effectiveness of driers are to be ensured	SCE/Unit Control Engineer
5.10	Laboratory check of hydrogen dew point is to be done once in a week on a portable dew point meter. In case the dew point is more than 0deg.C at casing pressure, the frequency of monitoring shall be increased to twice a week. The measured dew point on meter which is at atmospheric pressure has to be converted to dew point at casing pressure of 3.5Ksc by the graph. Dew point and absolute humidity table is given in annexure – III.	Asst/Dy. Mgr (Water lab)
5.11	Source of moisture shall be identified when dew point is reported as high and moisture level shall be brought down to safer value by hydrogen drier and purging. If necessary change over ineffective gas drier by stand by drier	SCE/Unit controller
5.12	It is preferable to maintain dew point at 0 deg. C or below at casing pressure. In case of difficulties in maintaining for short periods, the gas cooling system may be adjusted to get the cold gas temperature 20 deg. C above the dew point. While trying to achieve this, the manufacturer's limit of maximum cold gas temperature of 55 degC shall not be violated. Limits are given in annexure – II	SCE/Unit controller
5.13	Seal oil is potential source of moisture entry into generator casing and in case of increase in moisture in Hydrogen (dew point increase), turbine oil/ seal oil shall be checked for moisture content more frequently than the routine practice	Asst/Dy Manager (Oil lab)
5.14	MOT centrifuge availability shall be ensured to remove moisture from the oil	HOD(TMD)
5.15	H2 Drier availability shall be ensured for effective removal of moisture from the hydrogen	HOD(EMD)
5.16	All exceptions and deviations should be reported in maintenance planning meeting to dispense the deviations.	HOD(Chemistry)

7.0 REFERENCE DOCUMENT: OGN/OPS/SYST/009, Aug 1994

8.0 REVIEW:

GM (O&M) is responsible for reviewing the document on a 2-yearly basis and as required.

ANNEXURE – I

BASIC TERMS AND DEFINITIONS

A. WATER VAPOUR OR MOISTURE:

The mixture of air and water vapor at a given temperature is said to be saturated, when it contains maximum amount of vapour that it can hold. If the temperature of the mixture of air and water vapour is above the saturation temperature of the water vapour, the vapour is called as superheated and can be taken as gas for practical purposes.

B. ABSOLUTE HUMIDITY:

Absolute humidity is defined as the quantity of water vapour actually present in a given volume of air, without regard to its temperature. Thus, if air at any temperature contains 4 grams of water vapour per metre cube, its absolute humidity is expressed as 4 grams per cubic metre.

C. RELATIVE HUMIDITY:

Relative humidity is an expression in percentage terms of the degree of saturation of air at any given temperature. So, RH is equal to $100 \times \text{Absolute humidity} / \text{Saturation humidity}$.

Relative humidity depends on both the factors of temperature and moisture content. A variation in either alters the relative humidity.

D. DRY BULB TEMPERATURE:

The dry bulb temperature is the actual temperature of air measured by an ordinary thermometer.

E. WET BULB TEMPERATURE:

As the air is passed over bulb covered with wet wick, the moisture contained in the wick tends to evaporate. The cooling effect of evaporating lowers the temperature of bulb to a point corresponding to the rate at which the evaporation occurs. When the temperature measured by the wet bulb thermometer reaches steady state then the heat required for the evaporation of water vapour in the air is equal to the heat given out by the air by

convection to the thermometer. This states that the total heat of air leaving the wet bulb remains constant.

During the process of evaporation, sensible heat is only converted into latent heat of evaporation maintaining total heat of air constant and wet bulb temperature is the measure of total heat of air. Wet bulb temperature is a measure of degree of saturation or relative humidity. As relative humidity of air goes high wet bulb temperature approaches dry bulb temperature. At 100% RH wet bulb and dry bulb will show equal temperature.

F.DATLON'S LAW OF PARTIAL PRESSURE:

It states that the total pressure of mixture of gases is equal to the sum of the partial pressures exerted by each gas when it alone occupies the volume of mixture at the temperature of mixture.

G. DEW POINT:

The temperature at which the condensation of moisture in gas starts is known as Dew Point Temperature. In other words, the Dew Point of air/gas medium is the temperature at which the medium is fully saturated with moisture and the slightest drop in temperature will result in moisture appearance on the available solid surface. The Dew Point temperature increases with increase in pressure and thus a dew point of (-) 10 deg. C at atmospheric pressure corresponds to (+) 7.9 deg. C at 3.5 bar.

ANNEXURE-II

LIMITS OF HYDROGEN GAS TEMPERATURES AND DEW POINT IN DEG C

Sl.No	Parameter	Operation Value	Alarms	Protection of Generator
1.	Cold Gas Temperature	Normal:37 degC	High: 55 Very high: 60	More than 60
2.	Hot Gas Temperature	Normal:40-60 Max:<80	More than 80	More than 80
3	Temperature difference between cold primary water and H2 cold gas	More than 5	Low: 3 Very low: 1	
4	Primary water Temperature after cooler	Normal: 42 degC	55	At 55
5	Hydrogen Dew point at casing pressure	Less than zero, Zero or Atleast 20 deg C below cold gas temperature	High>0 degC Very High: 10 degC	

ANNEXURE – III

TABLE OF DEW POINT V/S ABSOLUTE HUMIDITY FOR HYDROGEN

Dew Point Temp. °C	Humidity Gm/m³	Dew Point Temp. °C	Humidity Gm/m³
-10	2.2	26	24.1
-5	3.2	27	25.0
0	4.8	28	27.0
1	5.2	29	28.8
2	5.6	30	30.4
3	5.9	31	32.0
4	6.4	32	33.8
5	6.8	33	35.8
6	7.3	34	37.7
7	7.7	35	39.6
8	8.3	36	41.6
9	8.8	37	43.6
10	9.4	38	45.8
11	10.0	39	48.3
12	10.7	40	50.3
13	11.3	41	53.4
14	12.1	42	56.1
15	12.8	43	58.9
16	13.6	44	61.9
17	14.5	45	65.0
18	15.4	46	68.2
19	16.3	47	71.5
20	17.3	48	75.0
21	18.3	49	78.6
22	19.4	50	82.3
23	20.5		
24	21.8		
25	23.0		

Hydrogen Dew point graph

(Atmospheric pressure Vs Operating pressure 3.5 KSC)

